Techno International New Town
Department of Computer Science & Engineering

Medical RAG System

Project-II Report (PROJ-CS781)

Presentation

Presented by : **Mind_Matrix**

Project Title : Medical RAG System
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PROBLEM STATEMENT

- The Challenge: Healthcare professionals struggle to access up-to-date medical protocols quickly
- Consequences: Delays in patient care, potential clinical errors, information overload
- The Need: Efficient, accurate, context-aware medical information retrieval at point of care
- Our Solution: AI-powered system combining retrieval and generation for instant medical knowledge access

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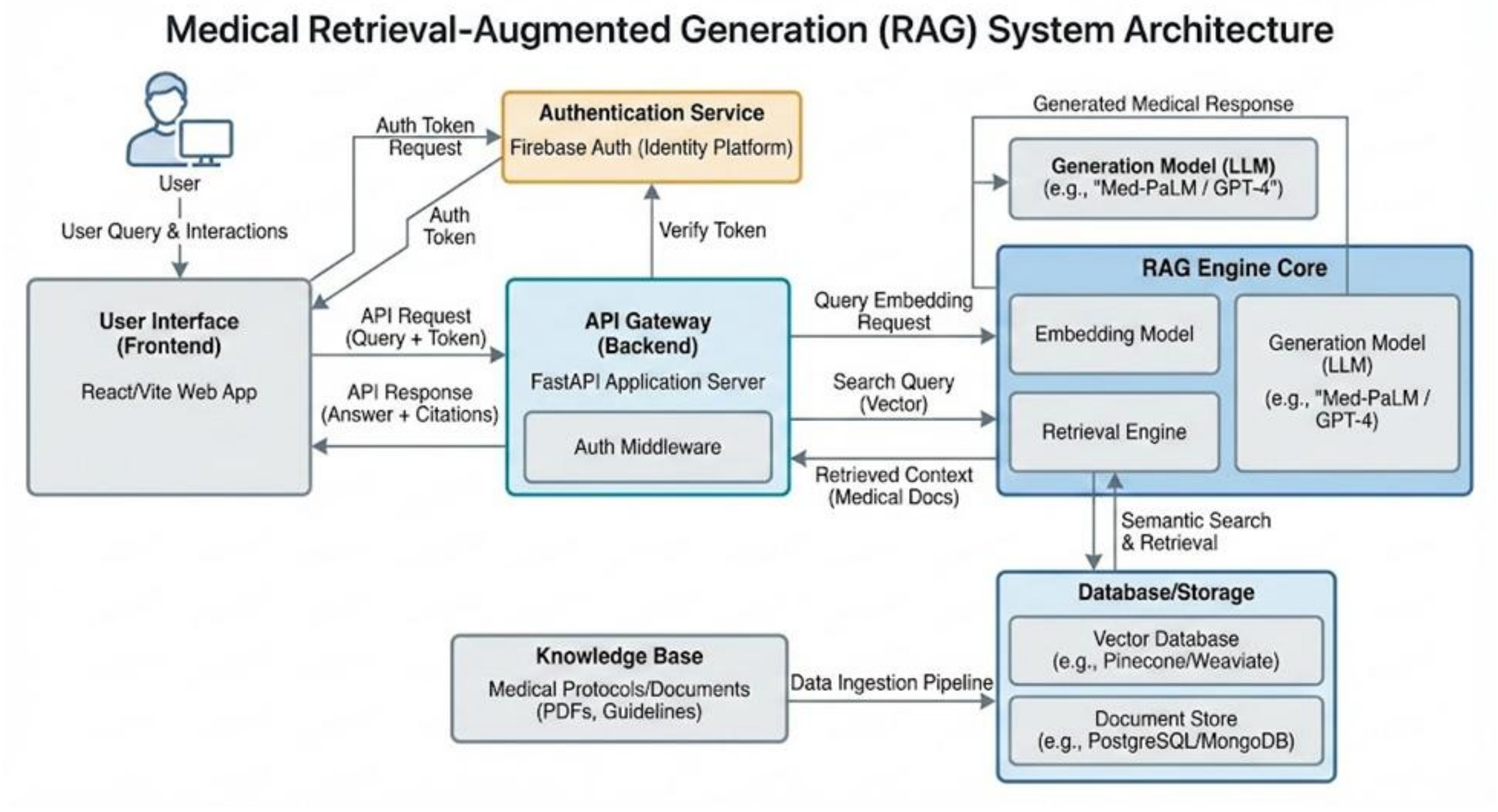
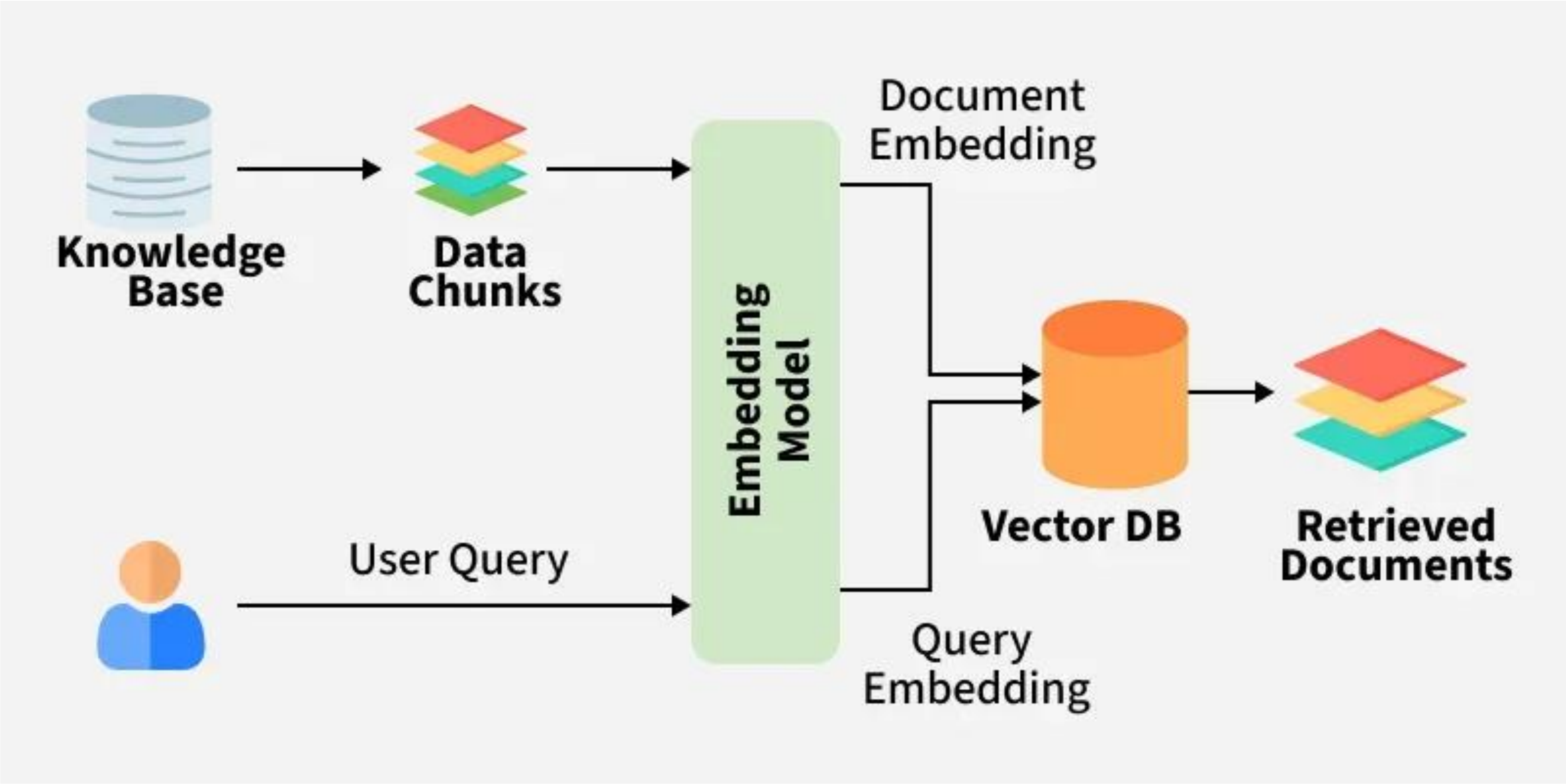
Technology Stack

Frontend: React + Vite for intuitive user interface
Backend: FastAPI for robust API management
Authentication: Firebase for secure user access
RAG Engine: Document embedding, retrieval, and AI generation
Knowledge Base: Curated medical protocols and guidelines
Design Principles: Modular, scalable, secure

System Architecture



What is RAG



- Retrieval-Augmented Generation (RAG) combines two powerful AI capabilities:
- Retrieval: Searches through knowledge base for relevant information
- Generation: Creates context-aware, natural language responses
- Why RAG? More accurate than pure AI, more intelligent than keyword search
- Result: Precise, evidence-based answers grounded in medical protocols

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KEY FEATURES

- *Intelligent Search: Context-aware medical query processing*
- *Secure Authentication: Firebase-powered user management*
- *Real-time Responses: Instant access to medical protocols*
- *Document Management: Easy addition of new medical references*
- *User-Friendly Interface: Intuitive design for healthcare professionals*
- *Accurate Results: AI-generated responses grounded in verified sources*

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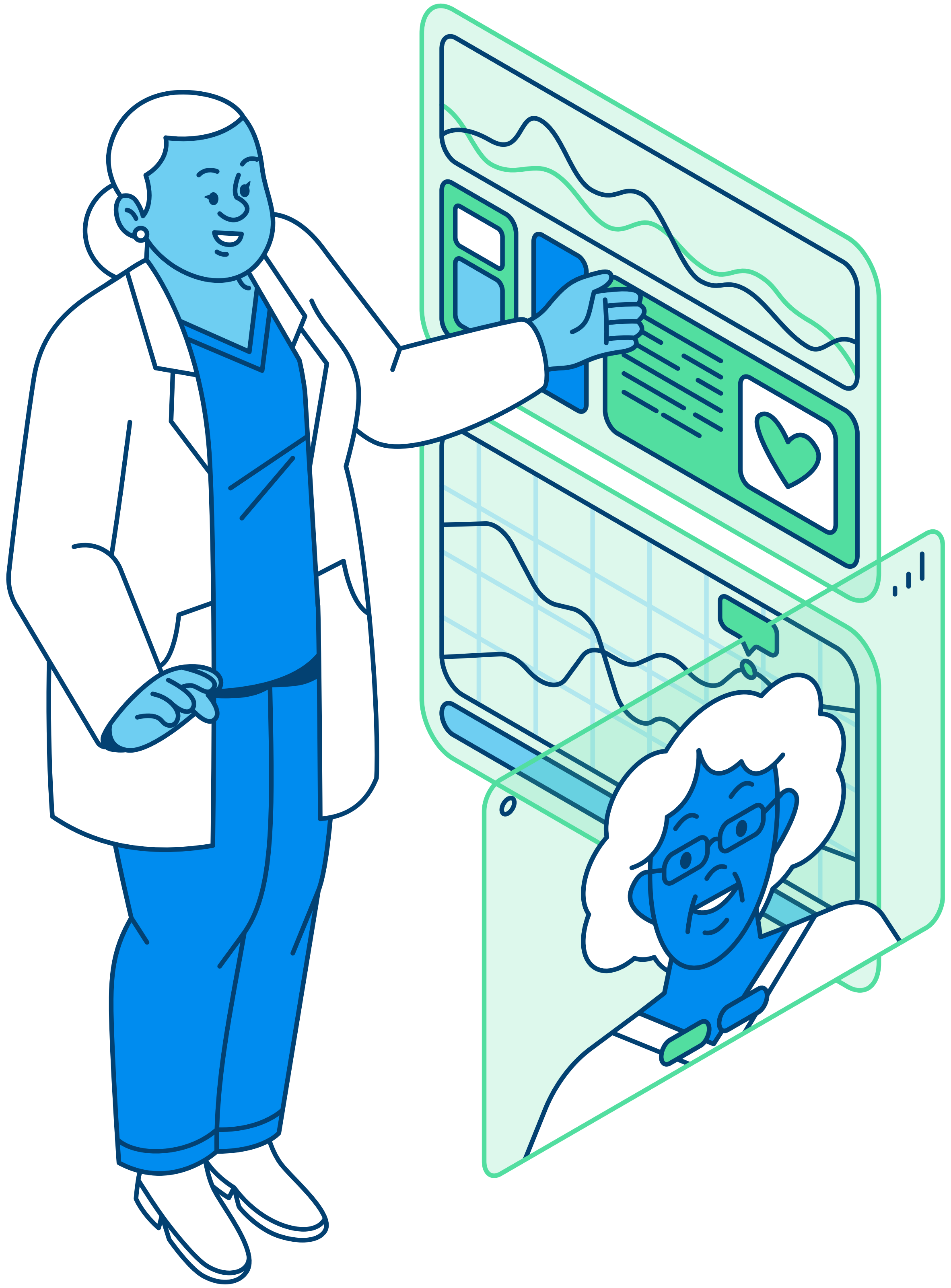
Use Cases & Benefits

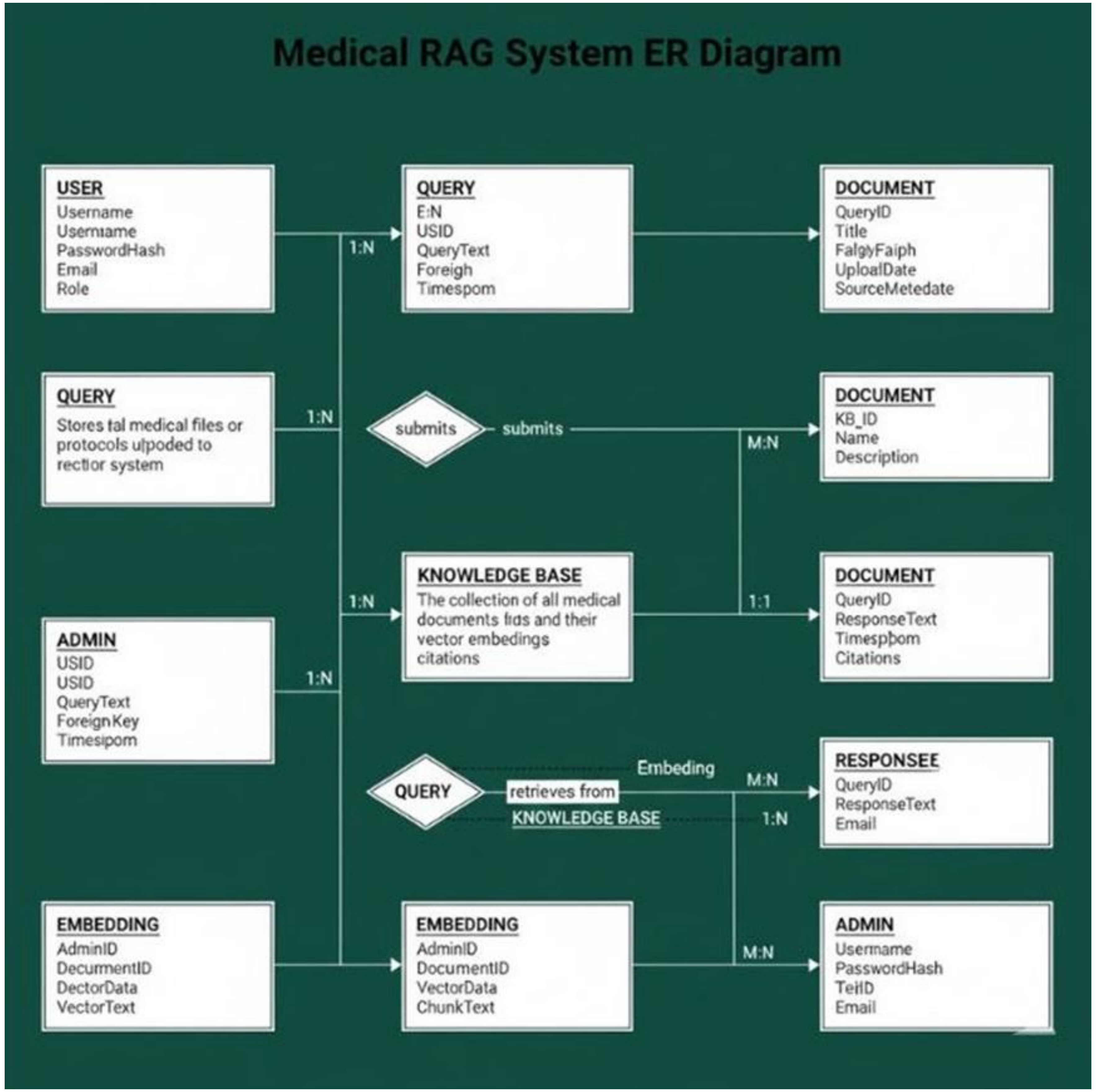


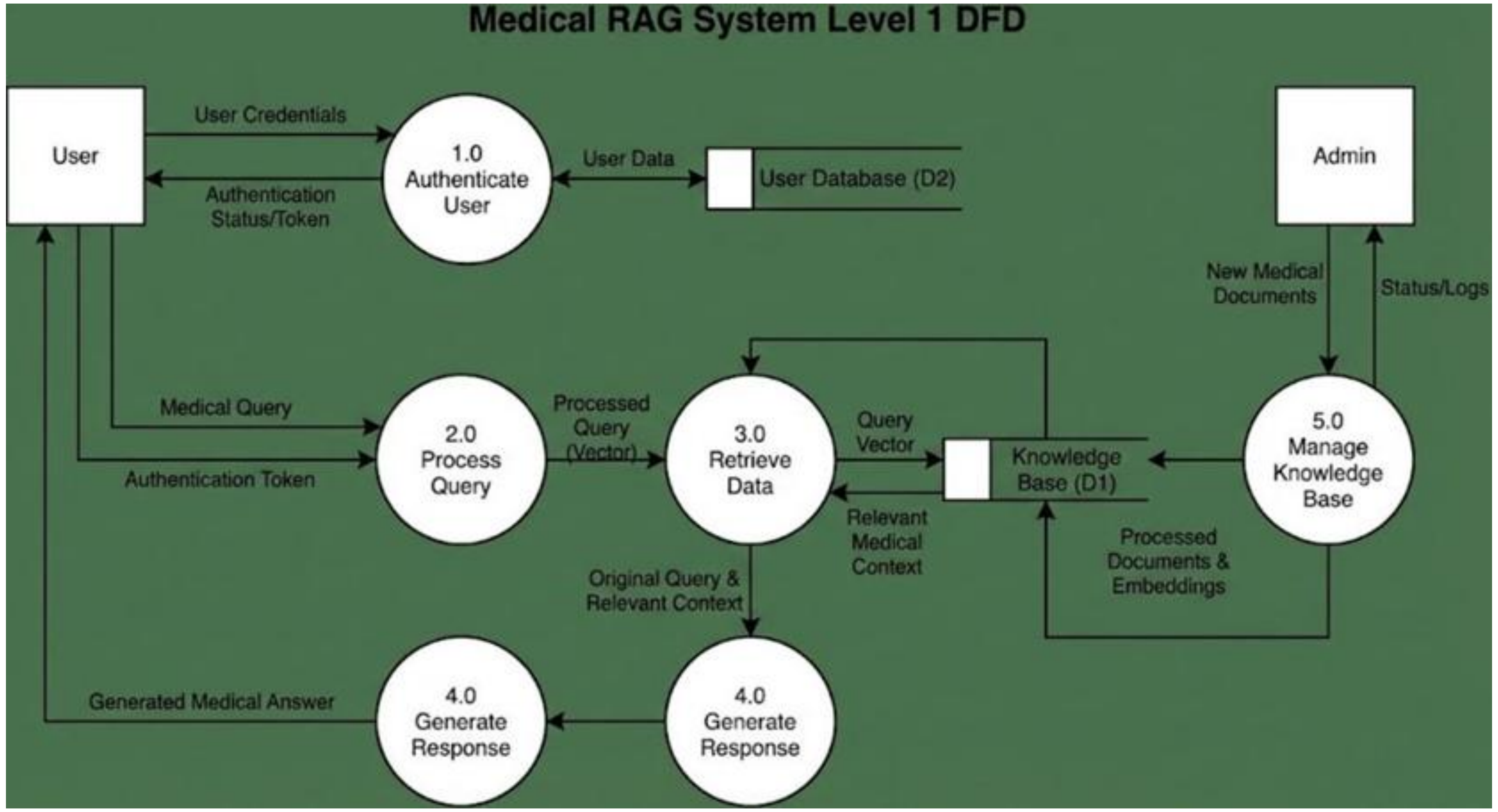
- *Primary Use Cases:Quick reference during patient consultations*
- *Medical protocol verification*
- *Treatment guideline lookup*
- *Clinical decision support*
- *Benefits:Reduced clinical errors*
- *Improved decision-making speed*
- *Enhanced patient care quality*
- *Time-efficient information access*

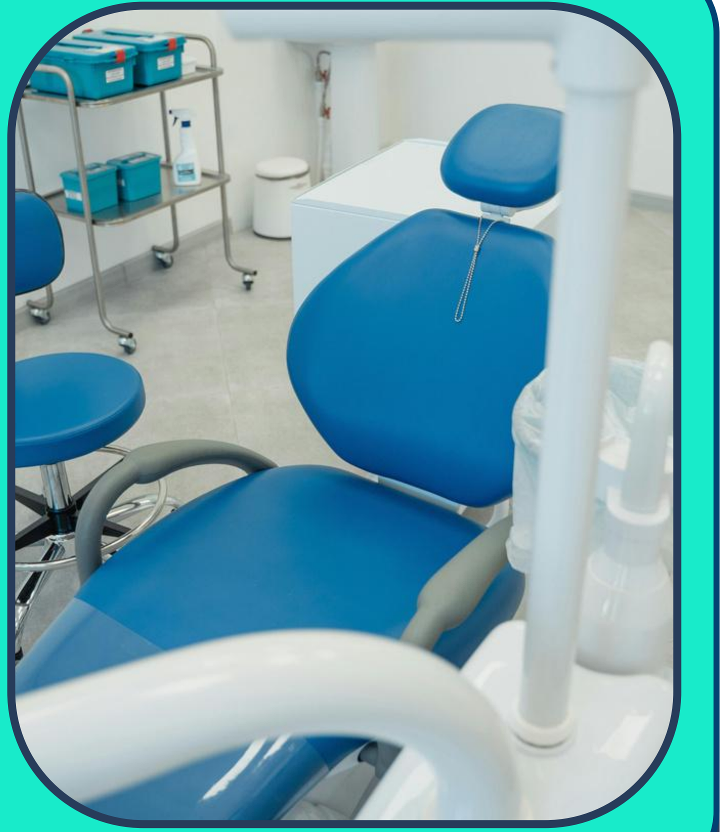
Future Scope

- Planned Enhancements:Integration with advanced LLMs (GPT-4, Claude)
- Multimedia knowledge base (images, videos, charts)
- Mobile applications (iOS & Android)
- Real-time team collaboration features
- EHR/HIS integration
- Advanced analytics dashboard
- Automated protocol updates

Database & Data Flow







Results & Impact

Achievements:

- Functional prototype with complete RAG implementation
- Secure authentication and user management
- Efficient document retrieval system
- Intuitive user interface

Expected Impact:

- Reduced clinical errors
- Faster decision-making
- Improved patient care
- Enhanced healthcare efficiency

Future Scope & Collaboration

- Mobile app development (iOS/Android)
- Multimedia content integration
- EHR/HIS system integration
- Real-time team collaboration features
- Advanced analytics dashboard

Conclusion: The Medical RAG System bridges AI and healthcare, providing healthcare professionals with rapid, accurate access to critical medical information

Thank You